

About the authors

The authors: Mr. Christian is a chemistry major, graduating this month; Dr. Berka (standing, right) an Associate Professor of Chemistry at Worcester (Mass.) Polytechnic Institute. They began research on polywater when Christian's curiosity was aroused by Simon Dresner's article, "Polywater . . . the Water That Isn't" [PS, Dec. '69] and have published several papers on polyliquids in scientific journals.

How You Can Grow Your Own Polywater

Some experts claim this rare substance doesn't exist. Yet here's how you can harvest enough of it for your own experiments

By P. A. CHRISTIAN and L. H. BERKA

Polywater, the laboratory curiosity that shook up the scientific world only a few years ago [PS, Dec. '69], can be prepared and studied by amateur scientists using simple techniques. Polywater is a mysterious "offspring" of normal water with dramatically different physical and chemical properties.

Thick and gooey, it refuses to freeze at 32 degrees F or boil at 212 degrees F. It is almost 50 percent denser than normal water and exhibits different optical properties.

The origin of polywater remains pretty much a mystery; recently its very existence has been contested.

Some scientists claim polywater is literally the sweat off the brow of those trying to produce it. They report that they have detected several constituents of human perspiration in samples of polywater tested with highly sensitive analytical instruments. Other researchers claim their samples contain large amounts of silicon in the form of silica gel or sodium silicate (water glass).

Both findings are still somewhat inconclusive since the greatest handicap in analyzing polywater is its rarity. It has been produced only in extremely minute amounts; most samples are visible only under a high-power microscope. In fact, the total world supply of polywater could be deposited in a thimble!

A number of methods have been used to prepare polywater; just why they work is not yet clear. Basically, all involve the condensation of water vapor in ultrafine Pyrex or quartz capillary tubes under controlled conditions. The method described here

is by far the simplest and, surprisingly, one of the most successful. We'll also describe test experiments you can perform.

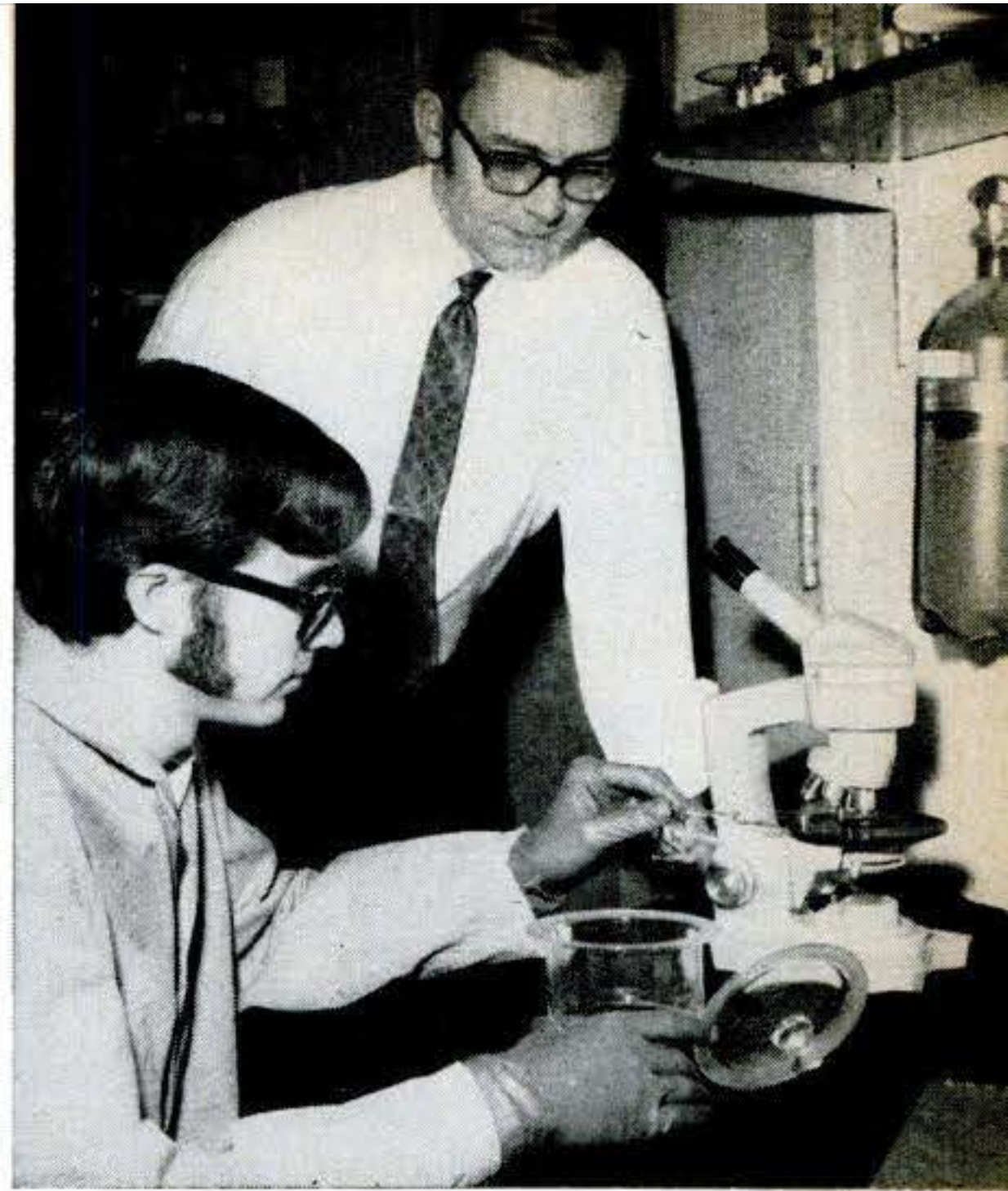
Materials you'll need. First, you must have access to a microscope with a magnification of at least 100. To prepare the capillaries, you'll need some sort of high-temperature torch; an oxyacetylene torch will do nicely. (Regular propane torches and blowtorches are not hot enough to melt Pyrex.)

The Pyrex tubing that has proven easiest to work with has a 0.5mm bore diameter and 5mm to 6mm outside diameter (it's often called thermometer tubing). You can obtain it from mail-order scientific-supply houses. You'll need at least 8' per batch of polywater. Some 1/4"-diameter solid Pyrex rods are also needed to make supports.

Another important piece of apparatus is probably sitting in your refrigerator: a snap-top, round, plastic refrigerator container of the type that soft margarine comes in. Also, for cleaning the tubing, get some ethyl (rubbing) alcohol or, ideally, some acetone.

You will also need fine nichrome wire (about 6") to bundle and suspend the drawn capillary tubes. A large Mason jar containing some calcium chloride (a drying agent often used in damp cellars) will serve as an airtight drying chamber. The nichrome wire, Mason jar, and calcium chloride should be available at your local hardware store.

Finally, you must have a supply of distilled water. You will probably be able to obtain some distilled



Authors study polywater sample in microscope. Christian wears rubber gloves to minimize chance of impurities. Inner white line in 400X view (bore of capillary) is granular-appearing polywater. *American Chemical Society copyright photo.*

water from your drugstore or gas station (where they use it to fill batteries). If not, try a local hospital or a bottled-water distributor to obtain a supply.

Clean the apparatus. Thoroughly wash the bore and outside surfaces of the Pyrex tubing by rinsing alternately with ethyl alcohol (or acetone) and distilled water. Plastic squeeze bottles with nozzles work very well for this. The last rinse should be with distilled water.

The snap-top refrigerator container, too, should be thoroughly washed with liquid detergent, then completely rinsed, with a final rinse of distilled water. Allow the container to air-dry, but protect it from dust. Also, try to avoid touching the inside surfaces of the container and cover.

Follow the instructions (box, next page) to prepare a sample. Then, to harvest your polywater, remove

Continued



How to prepare bundles of glass capillaries

Clean the Pyrex tubing as described in the article. Notch the tubes with a triangular file and break off sections at 4" intervals. Hold a tube in the flame (photo inset above) and heat a $\frac{1}{2}$ " segment in the middle until it glows. Remove the tube from the flame and draw it out rapidly and evenly to about 4' (above). Carefully snap off the unstretched ends.

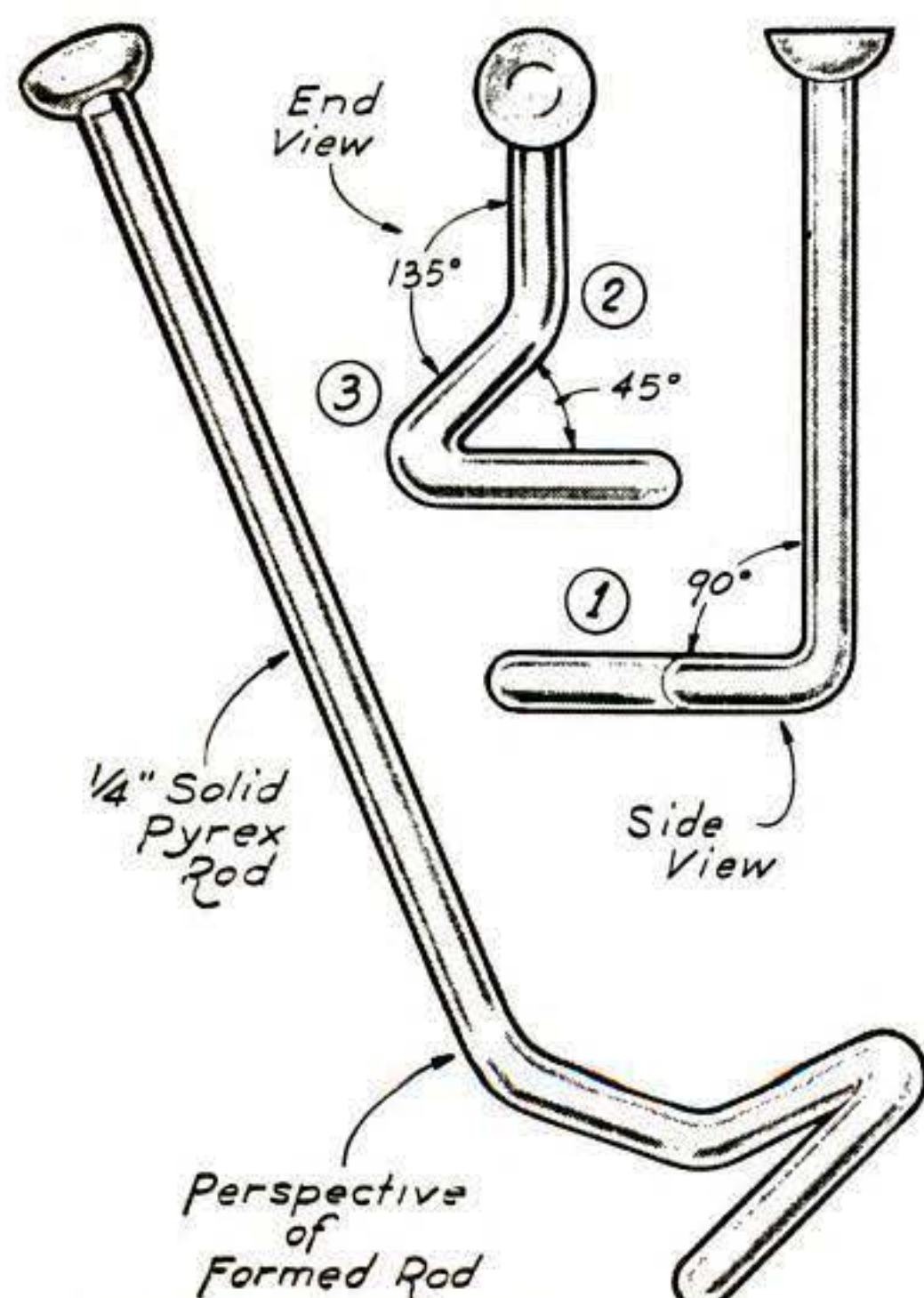
This filament should have a bore diameter of five to 20 microns (two to eight $\frac{1}{10,000}$ of an inch). Break the 4' filaments into 2" lengths. Each foot of undrawn tubing should provide you with 50 to 60 capillaries.

Capillary tubes must be suspended free from contact with the container walls, so a support must be prepared. Heat one end of a solid Pyrex rod and push it against a clean metal surface to flatten it (see drawings, left). Bending the rod is a little trickier. The bends should be made in the order shown and with the suggested angles.

Dimensions depend on your container's size. Some experimentation will be necessary to achieve the dimensions for a snug fit. The completed support should be washed in the same manner as the refrigerator container.

Using two lengths of nichrome wire, bundle the glass capillaries you've prepared securely together.

Attach the bundle to the support so that it swings freely about an inch below the rod. Add distilled water to the container to a depth of $\frac{1}{2}$ ". Position the bundle, snap on the cover, and set the container aside for at least a week.



Polywater—one of the

the bundle of capillaries from the support, place it in a small, clean, open pill vial and stand this vial upright in a Mason jar one-third filled with calcium chloride. Set aside the sealed jar for a week. This will eliminate any normal water present and concentrate the polywater.

After the capillaries have been dried by the calcium chloride, you must examine them individually under the microscope to locate the globules of polywater. Usually you can expect to find polywater in 10 to 30 percent of the capillaries.

Identifying polywater. A representative sample of polywater, as seen under the microscope, is shown—the white areas in the center of the narrow dark strip, the bore of the tube. A number of small granules appear to be dispersed throughout the material. This is characteristic of many polywater samples.

"Dry" polywater picks up normal water from the air, so you must store samples in a vial in the Mason jar. Polywater keeps well. Samples stored over calcium chloride for over a year show no visible change.

Experiments with polywater. You can easily investigate several of the unusual properties of polywater with your samples.

• **Heating behavior.** Place several capillaries containing polywater in a vial and heat them in your oven for five minutes at 550 degrees F. When you look at them under the microscope, you should still find some material in the tube. This demonstrates the very low volatility of polywater.

Also, you will probably notice that the column of polywater, which looked white or gray before heating, is yellowish and maybe even orange or dark brown after heating.

If you strongly heat a capillary in a test tube immersed in a sand bath (a can filled with beach sand) you will be able to "burn" the polywater. You can then observe the charred "remains" under the microscope. This discoloration and "burning" has been attributed to the thermal decomposition of various organic impurities present.

• **Cooling behavior.** Polywater has been observed to vitrify (become glass-like) at about -30 degrees F. Cool a capillary containing polywater and a capillary of equal diameter containing normal water to the lowest temperature possible in your freezer. Mount them side by side on a microscope slide.

Quickly remove the slide from the freezer and observe the two samples under the microscope in a dry room. The polywater should not appear

most interesting discoveries of the last decade—is still controversial

frozen, but the normal water will be frozen solid.

It may be necessary to coat the slide with some silicon anti-fogging compound used on eyeglasses to prevent fogging. If this is done, the capillary tubes should be sealed at both ends (using a match) to keep the silicon compound out of them.

● **Viscosity.** Polywater has been reported to be almost 15 times as viscous (thick) as normal water. Mount an open-ended capillary containing normal water in a larger tube using a drop of epoxy. Connect a large medicine-dropper bulb to the other end.

Now mount the assembly on the microscope stage so you can see the column of water. Squeeze the bulb slowly and observe the movement of the water column. Repeat with a capillary containing polywater. The polywater column should not budge.

● **Birefringence.** If you have a pair of Polaroid lenses from sunglasses, you've probably observed that if you place one lens on top of the other and rotate it, there is a certain position at which you can no longer see through them.

Certain materials, when placed between Polaroid filters in this opaque state, counteract the effect of the filter and appear bright. Such materials are called birefringent. The granules in polywater exhibit this behavior. Normal water is not birefringent.

To observe the birefringent granules in the polywater, first place one Polaroid lens in front of the microscope eyepiece. Then place the other

underneath a capillary tube containing a sample of polywater. Rotate the lens in front of the eyepiece to obtain maximum extinction of the light. The granules should be visible as bright splotches.

● **Solubility.** Polywater will dissolve in normal water. Place a capillary containing polywater on a microscope slide and put a drop of distilled water at each end of the tube. Under the microscope you will see the water diffuse into the tube and dissolve the column of polywater. A solution of polywater, called anomalous water, is the result. Anomalous water exhibits many of polywater's properties, but to a lesser extent, since it is diluted.

Other polyliquids? You can try making other polyliquids following the same procedure you used to produce polywater.

After the growing period, place the vial of capillaries in the open air (protected from dust) for a week prior to the week they stand in the Mason jar over calcium chloride. During this first week, any of the normal liquid present will be eliminated by evaporation. Since water vapor may have condensed out in the capillaries during this time, the second week is required.

Liquids chosen for experimentation should have low boiling points (about the same or lower than water) and be as pure as possible. **CAUTION: Many low-boiling liquids are highly flammable and their vapors often highly toxic. Most labels warn of these hazards. Such liquids should be**

used only in the absence of flame in a well-ventilated area.

Look for mixtures. Most liquids commonly available at drug and hardware stores are mixtures; the contents should be listed on the label. Mixtures pose a problem, however, since you don't know which component produced the residues in the capillaries.

If you feel venturesome, you might try trichloroethylene, a fairly pure form of which is sold in hardware stores in place of carbon tetrachloride for cleaning purposes. To our knowledge, no one has ever tried to make poly-trichloroethylene.

Other pure liquids, such as methyl (wood) alcohol, isopropyl alcohol, and glacial acetic acid, obtained from a chemical supply house, are likely candidates.

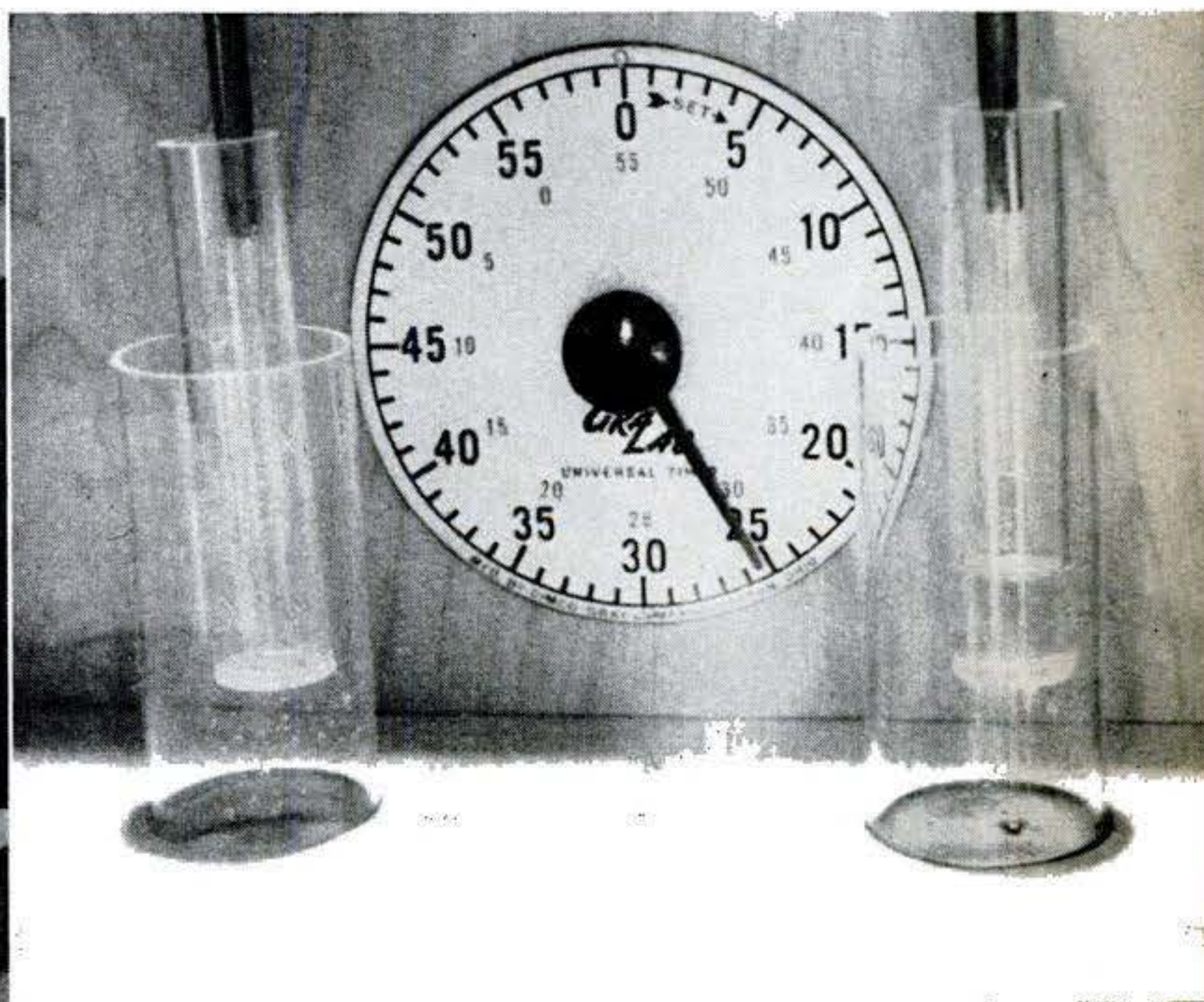
Two liquids. Another experiment that has not been tried is one in which columns are grown using a combination of two or more liquids in separate compartments in the container. Trichloroethylene and water could result in some interesting visual effects in the "poly-product" columns.

You can take photos of your samples, using the standard photomicroscopy technique. You might even blow up the photos to poster size.

Perhaps you can think of more things to do with your samples; we'd be glad to hear of your experiments with polywater. This substance, whatever it might be, is one of the most interesting scientific discoveries of the last decade—and still one of the most controversial. [E]



"Hands-on" exhibit at the Worcester Science Center is operated by authors—now more hirsute. Four-sided exhibit cube demonstrates polywater characteristics and apparatus.



Comparison exhibit displays viscosity differences of plain water, left, and liquid with polywater's viscosity. Viewer can time leakage through container holes.